Software Testing, Quality Assurance & Maintenance—Lecture 18

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More Dafny Motivation

Last time: talked about verifying Cedar (authorization library).

Dafny proves user-specified annotations and verifies absence of runtime errors.

Runtime errors: usually undefined behaviour at best, e.g. out-of-bounds array accesses, null dereferences, etc.

Alternative option: symbolic execution and bounded model checking. But Dafny is fully-static, and aims to work on real code.

Other Verified Software

- Paris Metro line 14 control system (B): 110,000 lines of models, 86,000 lines of Ada.
- seL4 (Haskell, Isabelle/HOL, C): conforms to its specification and maintains integrity and confidentiality.
- CompCert (Coq): a formally verified optimizing compiler
- IronClad (Dafny): establishes guarantees about how remote apps behave

First 3 by verification experts; IronClad by systems programmers.

All of these projects verified from the start.

More about formal verification & software development afterwards.

Dafny Examples

No slides; we'll work through

`https://dafny.org/dafny/OnlineTutorial/guide`